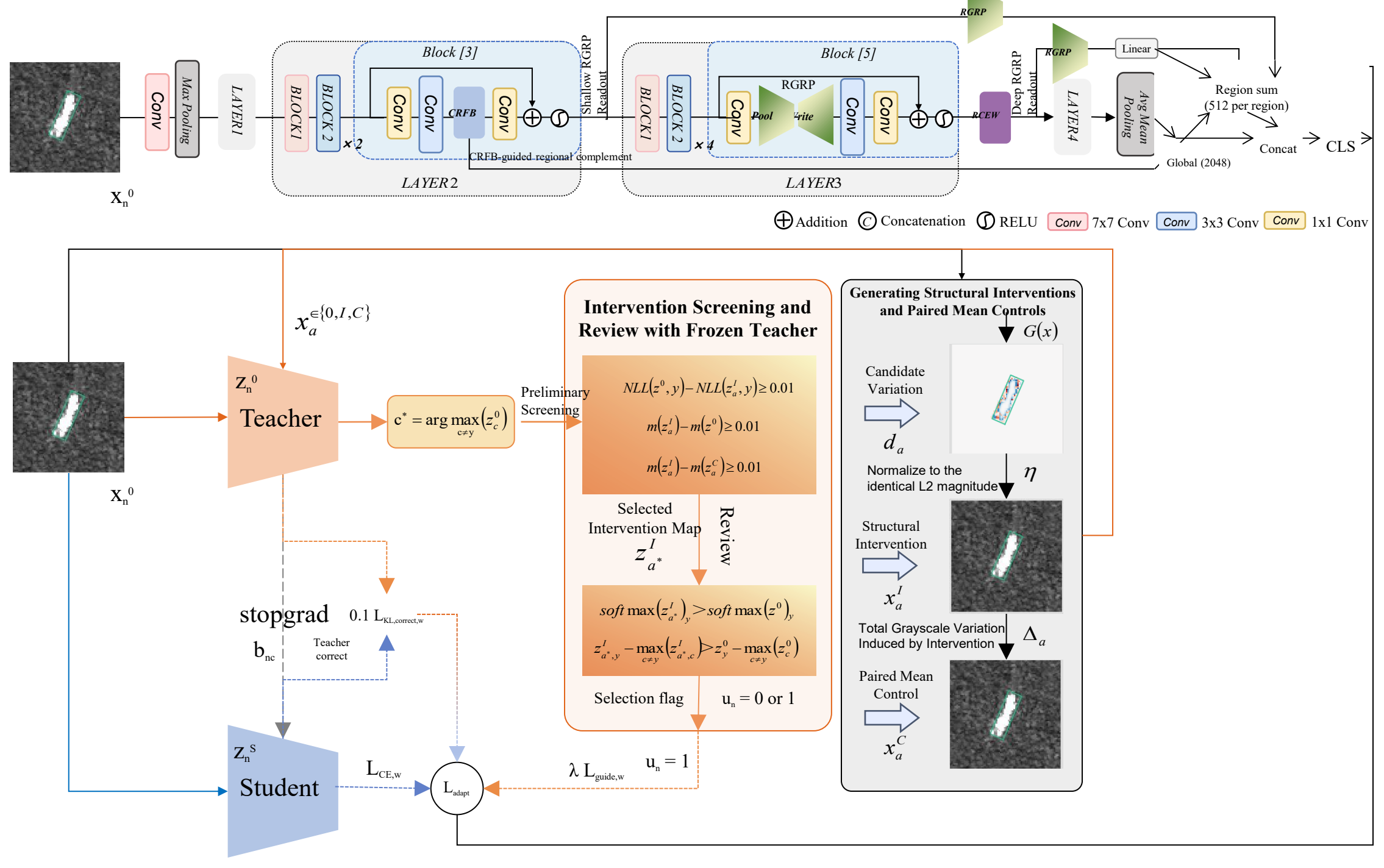

Conv
Max Pooling
LAYER1
BLOCK1
BLOCK 2
× 2
Block [3]
Conv
Conv
CRFB
Conv
Shallow RGRP Readout
CRFB-guided regional complement
LAYER2
BLOCK1
BLOCK 2
× 4
Block [5]
RGRP
Pool
Conv
Conv
Conv
RCEW
Deep RGRP Readout
RGRP
LAYER4
Linear
Avg Mean Pooling
Region sum (512 per region)
Concat
CLS
Global (2048)
Addition
Concatenation
RELU
7x7 Conv
3x3 Conv
1x1 Conv
Intervention Screening and Review with Frozen Teacher
Preliminary Screening
Selected Intervention Map
Review
Selection flag
Generating Structural Interventions and Paired Mean Controls
Candidate Variation
Normalize to the identical L2 magnitude
Structural Intervention
Total Grayscale Variation Induced by Intervention
Paired Mean Control
Teacher
Student
stopgrad
Teacher correct